

APPENDIX CE

PERFORMANCE-BASED DESIGN PROCEDURES FOR FIRE EFFECTS ON STRUCTURES

CE.1 SCOPE

Design approaches that consider fire effects on structures are generally categorized as either (1) standard fire resistance design (also referred to as the prescriptive approach) or (2) performance-based design (PBD). Although this appendix does not pertain to and should not be used for standard fire resistance design, it is discussed relative to PBD in this commentary.

Designers may elect to use PBD procedures for fire effects on a structure to meet stakeholder design objectives as permitted by Section 1.3.6 and alternative materials, design, and methods of construction and equipment provision in the building codes. When PBD procedures are used, the structure is typically designed for primary gravity and environmental loads and is then evaluated for fire exposure. An alternative material, design, or method of construction typically requires approval where the authority having jurisdiction finds that the proposed design is satisfactory and complies with the intent of the building code.

Structural fire resistance is the structure's ability to carry loads during exposure to fire conditions as well as provide a barrier to fire spread. Structural performance during fire exposure is often simply expressed as Fire Resistance > Fire Effects (Buchanan 2002). Three design philosophies are used for comparing fire resistance and fire effects, which are based on measures of time, temperature, and strength.

Time is used for standard fire resistance ratings in the building codes, where a particular configuration is shown, by testing or equivalent analysis, to provide adequate resistance to a standard fire exposure under test conditions for a period of time. *Temperature* is used in situations where it is postulated that a particular temperature will cause failure in a component or subsystem. A maximum allowable temperature is specified, and thermal protection is provided for defined fire exposures to ensure that the limiting temperature is not reached within the specified fire rating or period of time. *Strength* compares applied gravity loads and fire effects (e.g., thermally induced forces and deformations in structural members) to the fire resistance (e.g., temperature-dependent stiffness and strength) of the heated structural members and connections.

CE.2 DEFINITIONS

STANDARD FIRE RESISTANCE DESIGN: Standard fire resistance design methods are based on either the *time* or *temperature*. Fire resistance is most often defined as an hourly rating (e.g., 2-hour fire rated assembly) based on either results of the standard fire test or equivalent analysis methods.

Standard fire resistance testing provides a method of rank ordering through comparative testing of different structural assemblies under controlled laboratory conditions. Each test uses the same standard time–temperature curve, which continually

risks in temperature, to heat the structural members and assemblies with an established set of failure criteria. Standard time–temperature curves include ASTM E119 (2012) and ANSI/UL 263 (UL 2011b); international standards include ISO 834 (1999), CAN/ULC S101 (ULC 2007), and BS 476-20 (BSI 1987). There are also standard fire curves for hydrocarbon pool fires, such as ASTM E1529 (2010b) and UL 1709 (UL 2011a).

The standard furnace test has a standard time–temperature curve that provides severe heating conditions for test assemblies representative of field construction. Because of size limitations of furnaces, members and assembly sizes are limited. For instance, floor assemblies are typically tested at spans no greater than 17 ft (5.2 m), whereas an actual floor span may be much greater.

The fire rating of a member or assembly is based on the first failure criterion reached. For thermal response, there are temperature failure criteria measured by thermocouples applied to structural members. For flame and heat passage, the tested assembly cannot allow for ignition of cotton waste on the unexposed surface. For structural response, member deflections cannot become excessive (though excessive deflection is usually not explicitly defined in standard fire test methods). Fire resistance directories (e.g., *UL Fire Resistance Directory* 2013) provide a list of rated assemblies based on standard testing but do not provide information about the failure criterion upon which the listing is based.

When members or assemblies and their passive fire protection are similar to those already tested, methods to calculate equivalent performance for a standard fire test can be used to determine the fire resistance rating (e.g., *ASCE/SEI/SFPE 29-05* 2007). Analysis methods are available for structural members and fire barrier assemblies made from structural steel, plain concrete, reinforced concrete, timber and wood, concrete masonry, and clay masonry.

Standard fire resistance testing and equivalence computations do not include member connections, structural system response, or natural fire exposure. Standard fire resistance testing and equivalent analyses of hourly ratings do not provide the information needed to predict the actual performance of a structural system during structural design fires.

PERFORMANCE-BASED STRUCTURAL FIRE DESIGN: Performance-based structural fire design is useful for cases where standard fire resistance design would not address the design objectives of stakeholders. For example, a performance-based approach is appropriate for cases where performance of a structure during structural design fires needs to be quantified to properly assess risks to life safety and/or property protection. Building code variances for structural fire protection may require a performance-based approach to demonstrate the adequacy of an alternative design. Performance-based analyses also provide

opportunities to develop alternative designs that are optimized for aesthetics, functionality, and/or costs.

Acceptance of performance-based designs is subject to approval by the authority having jurisdiction. The designer must demonstrate that the design provides a level of safety that is equivalent or superior to that which would be attained by a design that conforms to the code's prescriptive criteria. Performance-based structural fire design provides a level of safety that is based on evaluation of structural system demand and capacity under fire conditions. Since the prescriptive approach is based solely on standard testing and does not consider structural system performance, the level of safety provided cannot be quantified. Consequently, there exists no practical method to quantitatively compare the level of safety provided by a PBD to that provided by the prescriptive approach. Therefore, it is the responsibility of the designer to properly demonstrate to the authority having jurisdiction that the PBD satisfies the required performance objectives and complies with the general intent of the building code.

The authority having jurisdiction should be contacted before initiating a PBD process to determine if their office is capable of and willing to support such alternative means, or if they have any concerns or requirements that need to be addressed. A peer review by an independent qualified party may be required by the authority having jurisdiction as part of the PBD process. Section 1.3.1.3 addresses PBD requirements for all types of load and performance requirements.

CE.3 GENERAL REQUIREMENTS

The frequency of major building fires is relatively low because of the small probability of ignition reaching flashover conditions (see Section CE.5.2 for a discussion of flashover). Occupant or fire department intervention and/or fire suppression system extinguishment of fire typically limit fire development before it becomes uncontrolled (Hall 2013). However, certain events and/or circumstances may result in uncontrolled fires that affect the structural system. In such cases, structural integrity should be maintained to ensure occupant life safety (see Section E.4.1). This assumes that the structural system is not significantly damaged by another hazard event, such as an earthquake or an explosion.

The term "fire effects" includes thermal response and corresponding structural deformations and loads induced by heating and cooling of structural systems during fire exposure, as well as temperature-dependent changes in structural stiffness and strength, nonlinear geometric and material responses, and restrained thermal expansion or contraction. All structural materials need to be evaluated for heating effects, but some materials, such as wood, may not experience additional strains or stresses from cooling effects. However, steel connections and fasteners used in wood construction may need to be evaluated for cooling. Fire effects may result in significant forces, rotations, deflections, and deformations of members and connections. Section loss resulting from fire exposure (e.g., because of spalling or charring) may also contribute to these effects.

PBD includes development of quantifiable performance objectives that are evaluated with appropriate analysis methods. The portion of the structure affected by fire, which includes members and connections as well as cooler surrounding sections that may provide restraint against thermal expansion, should be considered to determine the structural system performance and failure modes.

Analysis techniques used to evaluate fire effects on structures range in complexity from single element analyses to finite

element models that represent structural systems. Single members (columns or beams) and their connections can be analyzed in isolation for a structural design fire if reasonable assumptions can be made about restraint beyond the member and its connections, such as whether the rest of the structure provides rigid restraint to a heated member or whether the restraint of shear studs in a composite floor should be included or ignored.

Temperature-dependent strength and stiffness properties of materials may be based on peak temperatures for a structural design fire only if thermal expansion, restraint, and cooling effects can be conservatively neglected. Such considerations should include whether inelastic deformations, such as local buckling that is induced by restraint of thermal expansion, affect the member and connection behavior during cooling. For instance, bolt tear-out may occur during cooling of steel framing, which may result in member or subsystem failure. For composite floors, the thermal expansion of steel beams will be restrained as long as their shear stud connections remain intact.

Unlike single element analyses, finite element models of structural systems are able to capture the effects of thermal expansion, alternative load paths, secondary load-carrying mechanisms (e.g., catenary action), nonlinear material response, large displacement response, and connection performance. It is essential that the scope and complexity of the analysis techniques used to analyze structural response to structural design fires address the performance objectives in Section E.4 and are acceptable to the authority having jurisdiction.

The achievement of adequate performance and sufficient continuity and ductility for alternate load paths following member failure due to fire effects caused by a structural design fire requires consideration of connection capacity between structural members and application of the load combination in Eq. (2.5-1) with the appropriate resistance factors and member capacities for the given construction materials, as discussed in Section C2.5.

Since design and evaluation of structures for fire conditions is inherently multidisciplinary, multiple design professionals may be required. Design professionals may include structural engineers, fire protection engineers, architects, and others. Provided that multiple design professionals may be involved, the role and responsibility of each design professional should be clearly stated in contract documents.

For a standard fire resistance design, the architect usually serves as the responsible party for satisfying code requirements for structural fire protection. As such, the architect typically selects qualified fire resistance assemblies from available listings, perhaps with the consultation of fire protection engineers. For performance-based structural fire design, a team consisting of architects, fire protection engineers, and structural engineers is typically required. The fire protection engineer, or a design professional with similar qualifications, quantifies the fuel load, evaluates structural design fires, and estimates the temperature histories of structural systems. The structural engineer's primary responsibility is to evaluate the response of the structural system to fire effects based upon the provided temperature histories. The structural engineer may also assist the fire protection engineer in determining which structural systems should be evaluated for the structural design fires and in computing deformations of structural elements that may adversely affect the integrity of fire resistance rated assemblies, such as fire barriers.

CE.4 PERFORMANCE OBJECTIVES

Performance objectives primarily address structural stability and load path continuity, and requirements related to occupant egress.

Project-specific performance objectives may also need to be considered.

CE.4.1 Structural Integrity. Structural integrity supports life safety during fire in buildings and other structures. Accordingly, structural systems that support evacuation routes (e.g., corridors and exit stairs) and refuge areas should be evaluated for stability and load path continuity during structural design fires. Evaluation of stability and load path continuity should consider all supporting structural members and connections. For instance, a column under fire exposure may become unstable if lateral support elements lose their stiffness or load path continuity.

Stairways, horizontal exits, or even entire building floors may be designated as areas of refuge so that occupants can remain safely within the building during a fire. For instance, mobility-impaired occupants may need to remain within an area of refuge during a fire while awaiting rescue or evacuation assistance from emergency responders. Since areas of refuge are meant to serve as a place of safety according to the building codes, it is essential that structural systems that support these areas maintain stability, provide a continuous load path, and limit deformations throughout the heating and cooling of the structure under fire exposure.

Building codes limit egress travel distances to exits (e.g., stairways) but generally do not limit the total evacuation time. As the vertical remoteness of occupants from the point of discharge to a public way (e.g., a public street) is increased, the time required to evacuate the building increases. Hence, special consideration should be given to cases in which phased evacuation procedures are expected and longer occupant egress times are anticipated. For instance, in very tall buildings, occupants may be expected to remain on upper floors for hours, and even if those occupants are directed to use the stairways, the total evacuation may exceed one hour (SFPE 2013). In these cases, the structural performance of vertical exit stairways may represent a paramount concern to designers.

Building codes typically do not mandate how a building is to be evacuated, but there is often a requirement for certain buildings, such as tall buildings, to develop evacuation plans (SFPE 2013). Determination of the time frame necessary for occupant egress and the intended function of refuge areas generally requires the expertise of a fire protection engineer or a design professional with similar qualifications.

CE.4.2 Project-Specific Performance Objectives. In addition to the minimum requirements for structural integrity, project-specific performance objectives may be required. Project-specific performance objectives may address issues such as resilience aspects that consider recovery, property protection, business continuity, environmental protection, adequate structural support of fire resistance rated assemblies to limit fire and smoke spread, and/or structural support of ingress routes for first responders.

A greater level of structural performance may be required than that specified by Section E.4.1. For instance, the following example performance objectives may be applied to limit structural damage based on Risk Category:

- For buildings and other structures that meet Risk Category I criteria for low risk to human life in the event of failure, it may be necessary to avoid structural collapse from fire effects if the collapse is likely to damage valuable property within the building or surrounding properties, including other buildings and infrastructure systems.
- For buildings and other structures that meet Risk Category II or III criteria, or for any Risk Category that would likely damage surrounding properties if structural collapse

occurred, it may be necessary for the structure to endure structural design fires such that the primary structural system (e.g., columns, structural members having direct connections to columns, and lateral bracing members) remains stable with a continuous load path to supporting members during the heating and cooling of the structure. Damage to structural members or assemblies that do not compromise the stability of the primary structural system or continuity of the load path could be allowed.

- For buildings and other structures that meet Risk Category IV criteria, it may be necessary for the structure to endure structural design fires such that the entire structural system remains stable with a continuous load path to supporting members during the heating and cooling of the structure.

When designing to limit structural damage, buildings and other structures that meet Risk Category IV criteria may require that structural integrity be maintained for the entire structural system for structural design fires. By maintaining load path continuity and structural stability, enhanced property protection of adjacent areas of the building may be achieved, allowing for rapid reoccupation of areas not directly affected by fire exposure. For instance, if a critical facility experiences a severe fire in a given area, if structural integrity is maintained during and after the fire, such that there is no localized collapse and smoke damage and flame spread are contained, repair and recovery efforts will be limited primarily to the fire-affected areas.

Environmental protection objectives may include limiting the release and spread of hazardous or toxic chemicals to the air, ground and surfaces, or waterways because of loss of structural integrity.

It is desirable for fire-rated resistance assemblies to remain functional (resist fire spread and maintain adequate strength and stiffness for structural integrity) during structural design fires. Three limit states would need to be considered to evaluate fire resistance rated assemblies (e.g., fire barriers): (1) heat transmission leading to unacceptable rise of temperature on the unexposed surface, (2) breach of the barrier caused by loss of support, cracking, or loss of integrity, and (3) loss of load-bearing capacity. All three contribute to performance of fire resistance rated assemblies.

It is desirable for load-carrying elements (e.g., fire-rated floors and walls) that also serve as fire barriers to not have their fire resistance impaired because of deformations or other fire effects. When a fire resistance rated assembly is not load-bearing, the deformation of structural members supporting the assembly should not compromise its performance. However, criteria for limiting deformation of structural members supporting nonstructural fire resistance rated assemblies is not readily available. Fire resistance rated systems are qualified based on results of standard testing in which the supporting boundaries of the furnace (e.g., concrete floor) do not undergo deformation during heating. Addressing such performance objectives may require the designer to develop specific performance criteria per the discretion of the authority having jurisdiction.

CE.5 THERMAL ANALYSIS OF FIRE EFFECTS

Section E.5 provides requirements for determining structural design fires and the thermal response of structural members and connections. Structural design fires have the potential to affect the integrity and stability of a structure. Development of structural design fires involves consideration of compartment layout, boundary materials, ventilation, and fuel load that combine to create conditions that potentially threaten the structural system.

The designer should consider a sufficient number of structural design fires to properly address the risks with consideration of uncertainty associated with heating parameters. Based upon the time-dependent thermal boundary conditions from fire exposure, the thermal response of structural members and connections can be determined based upon principles of heat transfer.

CE.5.1 Fuel Load. Structural design fires depend on the fuel load and its distribution. Fuel load is commonly expressed as a fuel load density, or fuel load per unit floor area. NFPA 557 (2012b) establishes a basis for selecting fuel load density and distribution. Other methods may be used if acceptable to the authority having jurisdiction. Although the fuel load density concept implies a uniform distribution of combustibles in compartments, the actual distribution of combustibles may need to be addressed for analyses of structural design fires where localized heating effects may be significant.

The fuel load based on the contents of a building, space, or area typically varies because combustible materials release different levels of thermal energy when burning. For instance, plastics generally release more energy per unit mass than wood products. The conversion of building contents into an equivalent mass based upon their potential energy provides a consistent basis for determining the total energy of the fuel load. The fuel load in equivalent mass measured in pound mass (lbm) (kilograms (kg)) can be readily converted into total energy measured in British thermal units (Btu) (kilojoules (kJ)) for use in characterization of structural design fires.

CE.5.2 Structural Design Fires. Structural design fires are structurally significant fires based on the physical parameters, such as building layout, compartment boundary materials (e.g., walls), ventilation openings (e.g., doors and windows), and fuel load that are specific to a particular building space or spaces. Structurally significant fires include those that are not controlled by active fire protection systems, such as automatic fire sprinklers or firefighting efforts. Other design fires considered for fire detection, evacuation, or other fire-related issues may not be structurally significant fires.

Provided that there is sufficient oxygen available to support combustion, the duration of a structural design fire is dependent upon the heat release rate history of the fire and the total energy of the fuel load. A structural design fire reaches burnout when either the available fuel load is fully consumed or there is insufficient oxygen to support combustion. The materials involved in a fire significantly affect the heat release rate history. Hence, the heat release rate history of a fire is not necessarily correlated with the total energy of the fuel load.

Structural design fires should be evaluated using methods acceptable to the authority having jurisdiction. In certain cases, it may be necessary to perform fire modeling to evaluate structural design fires. Fire modeling generally requires the expertise of a fire protection engineer or a design professional with similar qualifications. Most fire models simulate the effects of fire (e.g., hot air and smoke flows) and not the phenomena of combustion and flame spread. Where fire modeling is required to determine time-dependent thermal boundary conditions on the structural system, the designer should substantiate the model according to SFPE G.06 (2011a).

Based upon the fuel load, ignition(s), and arrangement of compartments and ventilation openings, a structural design fire may be broadly defined as one of the following types: enclosure fire, localized fire, exterior fire, or traveling fire. SFPE S.01 (2011b) provides methods to determine time-dependent thermal boundary conditions on a structural system caused by a structural

design fire for either enclosure or localized fires. NFPA 80A (2012a) provides similar methods for exterior fires.

Events such as an earthquake or flood may result in enclosure, localized, and/or traveling fires within a building. Postevent fires may have compounding factors such as dispersed flammable contents, electrical malfunctions, interrupted power and/or water supplies, damage to fire sprinkler systems, or overextended emergency responders. Also, structural damage caused by a severe event may be exacerbated by a fire (e.g., reinforcing steel exposed by concrete spalling at a connection or loss of member insulating materials).

Enclosure Fire. An enclosure fire is affected by the compartment(s) in which it is contained. As a hot upper gas layer forms with the progression of fire, it reradiates heat back to the fire and fuel packages. The compartment boundaries may radiate inward as well. These conditions may eventually lead to flashover, at which point the fire is considered to be a fully developed fire. Flashover occurs when there is a rapid transition from localized burning to simultaneous burning of all combustible materials within the enclosure. Flashover can only occur in an enclosed compartment with sufficient fuel and ventilation, where the ceiling can trap hot gases that lead to radiant heating of all fuels to the point of combustion.

In most fires, there is variability in the fire size, depending on which items are ignited first and how the fire grows and spreads. Focusing on fully developed fires eliminates much of this variability since a fully developed (postflashover) fire is less sensitive to which items are ignited first and how the fire grows. For PBD analyses, neglecting the heating of the structure during the growth stage of a fire is usually a reasonable assumption since heating of the structure during the fully developed stage is much greater than heating during the growth stage.

Localized Fire. A localized fire burns combustibles at a given location and does not reach flashover. Localized burning occurs in open exposures, large spaces, areas with high ceilings, or other locations that are not conducive to flashover. This typically occurs in relatively large compartments or spaces, when the fuel is concentrated within a region. Fires that do not reach flashover may produce localized heating on the structure.

Exterior Fire. Exterior fires may lead to ignition and subsequent fire exposure within a building, possibly on multiple floors. For example, buildings in close proximity may mutually increase the risk of a large fire exposure through heat exchange between buildings. Flame impingement and convective heat transfer from exterior fires, sometimes from fires extending out windows from lower floors, may also create a fire hazard.

Traveling Fire. A traveling fire is characterized by the spread of fire from combustible to combustible across an open plan that does not burn simultaneously throughout the entire compartment. These fires move across areas as flames spread, burning over a limited area at any one time. Traveling fires are characterized by areas with combustibles that are not yet burning, a fire front with generally intense heating, and a trailing region where fuels have been largely consumed.

CE.5.3 Heat Transfer Analysis. The thermal response of the structural system depends on the structural design fire and the three modes of heat transfer: conduction, convection, and radiation. All three modes of heat transfer typically occur when a structure is heated by fire.

Heat transfer analysis methods are specific to the material's physical and chemical response to heat and are used for homogeneous and nonhomogeneous materials. Material responses may include charring, intumescenting, dehydration, phase changes, and chemical reactions. These material responses and associated

properties may significantly affect how heat transfer analyses are conducted.

Relevant material thermal properties for heat transfer analyses include density, thermal conductivity, emissivity for exposed surfaces, and specific heat (which may include heat effects caused by phase change, if any). Many of these material properties have strong temperature dependence. Sources with temperature-dependent thermal properties for steel, concrete, masonry, and timber are listed here:

- ACI/TMS. (2007). *Code Requirements for Determining Fire Resistance of Concrete and Masonry Construction Assemblies*, ACI 216.1-07/TMS-216-07.
- American Wood Council (AWC). (2015). *National Design Specification (NDS) for Wood Construction*, AWC NDS-2015.
- European Committee for Standardisation. (2004a). *Eurocode 2: Design of Concrete Structures. Part 1-2: General Rules—Structural Fire Design*, EN 1992-1-2.
- European Committee for Standardisation. (2005). *Eurocode 3: Design of Steel Structures. Part 1-2: General Rules—Structural Fire Design*, EN 1993-1-2.
- European Committee for Standardisation. (2004b). *Eurocode 5: Design of Timber Structures. Part 1-2: General—Structural Fire Design*, EN 1995-1-1.
- European Committee for Standardisation. (2003). *Eurocode 6: Design of Masonry Structures. Part 1-2: General Rules—Structural Fire Design*, EN 1996-1-2.
- Forest Products Laboratory (FPL). (2010). *Wood Handbook: Wood as an Engineering Material*, Forest Products Laboratory General Technical Report FPL-GTR-190.
- Kodur, V., and Harmathy, T. (2008) *Properties of Building Materials*, SFPE Handbook of Fire Protection Engineering.

For design purposes, constant values of thermal properties can be used if they yield conservative results. Depending on the heating or cooling conditions at exposed surfaces, applicable values for convection heat transfer coefficient should be used. Thermal insulation should be analyzed using the specified minimum thickness.

Heat transfer analyses inherently assume that the materials remain in place during the fire exposure. If insulating materials are expected to fail during a structural design fire, either the heat transfer analyses should account for the resulting increased heating of the structure or the insulation design should be modified. The deformations of structural members during structural design fires may need to be considered as part of the evaluation of mechanical integrity of fire resistance rated assemblies.

The temperature histories of structural members and connections comprising the structural system should be determined using heat transfer analyses as permitted in SFPE S.02 (2014). Other approved methods may be used if acceptable to the authority having jurisdiction.

CE.6 STRUCTURAL ANALYSIS OF FIRE EFFECTS

Structural analysis of fire effects requires consideration of the heated members and connections with consideration of the entire structural system. A single member analysis may be justified where only a single member is affected by a fire without consequential effects from surrounding members. A systems approach requires consideration of thermal expansion of heated sections and restraint by cooler adjacent framing, thermally induced forces and displacements on connections, the response

of floor systems, and thermally induced failure modes across the structure (McAllister et al 2013). Structural elements may have large deflections that are an order of magnitude greater than deflection limits normally anticipated for structures (McAllister et al. 2012). Large deflections may induce forces in adjacent structural assemblies (e.g., members and connections).

Thermal Expansion and Restraint. Floor systems may experience thermal restraint during heating from columns and cooler adjacent floor members; interior bays typically experience more thermal restraint than exterior bays. Columns, on the other hand, typically do not experience significant thermal restraint from floor systems. However, if there is lateral bracing in place, it may impose some thermal restraint on the braced column section, depending on the framing geometry and member temperatures, if the bracing members and adjacent column are at a significantly lower temperature.

A temperature gradient through the depth or thickness of a structural element causes differential thermal expansion between the hotter and cooler external surfaces. Differential expansion results in curvature for simply supported members. For members with partial or full rotational end restraint, a temperature gradient results in a strain gradient through the depth of the member because of thermal restraint.

The effect of thermal expansion and contraction needs to be carefully considered. Thermal expansion and contraction of construction materials may generate forces sufficient to cause yielding or fracture, depending on the temperature reached and the degree of restraint provided by the surrounding structural system to the thermally induced actions (Gillie et al. 2002). In fact, thermal restraint may dominate the behavior of framing systems, particularly floor systems, with degradation of stiffness and strength a secondary factor (Bailey et al. 1999). Fire-exposed elements that have experienced plastic deformations caused by weakening and thermal restraint may experience tensile strain as the structure cools and may induce forces in adjacent structural assemblies (e.g., connections), depending on the level of thermal restraint.

Columns. Fire effects on steel columns include loss of strength and stiffness, thermal expansion, and P-delta effects under thermal gradients, which may affect global and local column buckling strength. Design procedures for fire effects on steel compression members are provided in Appendix 4 of AISC (2010). The equations are based on analyses conducted by Takagi and Deierlein (2007). The effects of thermal gradients on the axial-moment capacity of steel wide-flange sections are discussed by Garlock and Quiel (2008). Analyses by Seif and McAllister (2013) indicate when elevated temperatures may result in local and global buckling modes of steel wide-flange sections.

Fire effects on concrete columns include loss of strength and stiffness in both the concrete and reinforcement. The primary causes of fire damage to concrete are deterioration in mechanical properties of the cement paste and aggregate, cracking, and spalling (Khouri 2001). Spalling occurs in normal-weight concrete as well as high-strength concrete (Hertz 2003). Concrete cover serves as insulation for reinforcement, so concrete cracking or spalling allows direct heating of reinforcement. Lie and Irwin (1993) provide models based on experimental data for predicting the performance of reinforced concrete columns with rectangular sections, based on axial deformations and temperature through the concrete section. Kodur and McGrath (2003) present experimental results for high-strength concrete columns that include the effects of concrete materials, loading, and spacing of ties. Kodur and Phan (2007) describe the factors that influence spalling in high-strength concrete members.

Floor Systems. A study of steel beams and composite floor systems exposed to a range of heating scenarios (Moss et al. 2004) found that the system behavior at elevated temperatures caused interrelated changes in the deflected shape, axial force, bending moments, and internal stresses that varied with the type of support condition and thermal restraint.

Composite floor systems tested at Cardington (British Steel 1999) and for the FRACOF (Zanon et al. 2011) and COSSFIRE (Zhao and Roosefid 2011) programs found that composite floors with beam lengths less than 30 ft (9 m) did not experience failures during the heating phase, but connection failures did occur during the cooling phase if significant deformation occurred in the floor beams during heating. Wang et al. (2011) found that connection types and axial restraint for floor beams influenced the response of floor systems, primarily during the cooling phase. In contrast, the WTC 7 numerical analysis of composite floors predicted that shear stud and beam connections failed during the heating phase of long-span composite floor systems with one-sided beam-to-girder floor beams of 50- to 56-ft (15- to 17-m) length (McAllister et al. 2012). This numerical prediction was not verified through observation or physical fire testing because of the large size of the floor bays and inability to scale structural responses to fire. Shear stud failure has not been observed in any fire event or in any structural fire tests. The lack of observations may be caused by the limitations of full-scale structural fire tests to date. In structural fire tests, floor beam length is typically less than 20–30 ft (7–10 m), columns and connections are often protected from heating, and/or substantial restraint of the steel beam and concrete slab prevents thermal expansion of the composite section. Failure modes for many structural fire tests with shorter, restrained floor beams tend to occur during the cooling phase at the beam-to-column connection (Bisby et al. 2013). Bailey et al. (1999) and Elghazouli and Izzuddin (2001) assessed the performance of the Cardington floor systems and identified response mechanisms of the floor system to heating and cooling effects. Bailey (2004) presents a performance-based design approach that considers membrane action in the composite floor system.

Reinforced concrete floors, including cast-in-place and precast/prestressed construction, are typically designed by providing a specified cover thickness over reinforcement. In general, there is little guidance for the PBD of concrete floor systems for structural design fires. However, PCI (2011) provides design guidance for precast and prestressed concrete floor systems. Whereas heating of reinforcement is to be avoided, the heating of prestressed strands in floor systems should be of particular concern, as loss of prestress may significantly degrade the floor system performance.

Floor Connections. The performance of connections needs to be considered in the structural analysis of fire effects, particularly connections in floor systems. For instance, steel shear connections may experience bolt shear, local buckling or tear-out of connection plates, or local buckling of the beam flange near the connection. Parametric studies of single plate shear (fin) connections (Yu et al. 2009; Selamet and Garlock 2010; Hu and Engelhardt 2011) and double angle connections (Pakala et al. 2012) identified critical dimensions and component interactions that control connection behavior at elevated temperatures. Huang et al. (1999) evaluated the role of shear stud connections by comparing models with varying levels of composite action against test data from the Cardington fire tests of composite beams.

Heating of concrete and reinforcement in concrete framing (cast-in-place or precast) may result in concrete spalling that

accelerates reinforcement heating and loss of strength in the reinforcement.

Failure Modes. Fire-induced failure modes include large deflections, member buckling (local, global, or lateral torsional), connection failures (bearing, bolt tear-out, bolt shear, or weld failure), reinforcement and anchorage failures, and section loss (concrete spalling, cracking, or crushing or charred timber sections).

As temperatures increase sufficiently to reduce the strength and stiffness of the cross section, yielding or buckling modes may occur at arbitrary point-in-time (service) load levels. If such weakening occurs for members with temperature gradients, the resulting gradient in member stiffness and strength can alter the combined axial load and moment resistance of the member (Garlock and Quiel 2008). The strength and stiffness gradient may also cause the member centroid (i.e., center of strength and stiffness) to shift toward the cooler (i.e., stronger) side of the cross section. This centroidal shift induces moments in axially loaded elements.

CE.6.1 Temperature History for Structural Members and Connections. Temperature histories of structural members and connections depend on the thermal response for structural design fires. Temperature histories may include thermal gradients across a section or along a member length.

Thermal finite element analyses typically use 2D or 3D models with a fine mesh of solid elements (element size on the order of inches (centimeters)), whereas structural analyses typically use a coarser mesh of shell and/or beam elements (element sizes on the order of feet (meters)). Careful consideration should be given to the tradeoff between optimal model features for each analysis versus mapping of temperatures between two sets of nodal data. For example, a simplistic heat transfer model, such as the lumped mass method, provides uniform temperatures; uniform temperatures may be inappropriate for a structural system that is likely to experience significant temperature gradients. There may be situations where it is appropriate to use the same mesh discretization in the thermal and structural models, so that transfer of nodal temperature data is seamless.

In most cases, results from heat transfer models are mapped to significantly fewer nodes of beam and shell elements in structural models. The average rate of temperature changes in structural members and connections typically occurs on the order of minutes, and temperature data sets for the structural system can be input at set intervals to reflect the progress of heating and cooling. Temperatures may be interpolated linearly between the data sets during the structural analysis.

CE.6.2 Temperature-Dependent Properties. At elevated temperatures, the strength and stiffness of the material(s) comprising a structural assembly change. Sources for strength and stiffness degradation for steel (including prestressing steel), concrete, masonry, and timber are listed here:

- American Concrete Institute. (ACI). (2007). *Code Requirements for Determining Fire Resistance of Concrete and Masonry Construction Assemblies*, ACI/TMS.
- ACI. (2001). *Guide for Determining the Fire Endurance of Concrete Elements* (Reapproved 2001), 216R-89.
- American Institute of Steel Construction. (AISC). (2010). *Specification for Structural Steel Buildings*, 14th Ed., Appendix A4, “Structural Design for Fire Conditions.”
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The Eurocode defines separate curves for material properties at elevated temperatures for the proportional limit and the yield stress. However, the yield stress at elevated temperatures is often defined at a 0.02 strain in the Eurocode. For ambient temperatures, the yield stress is defined at a 0.002 offset strain (other methods are also permitted as defined in ASTM A6/A6M (2010a), and the proportional limit is not defined by an ASTM test method. However, values for the proportional limit are given in AISC (2010) Appendix 4 in terms of the ratio to the yield strength. The analyst should be careful to note these distinctions in material models.

Where possible, temperature-dependent material properties should be obtained from consensus standards. Alternative sources of data, such as research studies and independent tests, may provide useful data. However, reliance upon a single test or source of data should be avoided. At a minimum, test data from several sources should be collected and used to develop a representative set of temperature-dependent material properties.

CE.6.3 Load Combinations. For extraordinary events, such as structurally significant fires, load combinations 2.5-1 and 2.5-2 are used to evaluate the structural system performance. This load combination was developed for extraordinary events that may lead to ultimate limit states such as gross inelastic deformation or partial collapse.

Load combination 2.5-1 is used to perform a safety check on a structure designed for the basic load combinations at room temperature conditions and to evaluate the effect of elevated temperatures. The force in structural members caused by fire effects, A_k , has a load factor of 1.0 (Ellingwood and Corotis 1991; Ellingwood 2005). The live load factor of 0.5 is intended for typical occupancies and arbitrary point-in-time live loads that likely exist during a significant fire. The 0.5 live load factor is also used in other load combinations in Section 2.3 when live load is a companion load and not the principal load. Note that the live load in this load combination differs from the approach used in standard fire test methods, where the assembly is loaded to its design limit for member stress during the standard fire exposure, which represents the application of the full dead and live load.

Whereas gravity loads for the structure remain constant during most fires (assuming that most of the building contents are not burning), time-dependent temperature histories may result in time-varying member strength and thermally induced forces, depending on the temperatures reached by structural members.

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